

Resource Profiling of Agentic LLM Pipelines: Quantifying Async Speedup and Framework Overhead in PocketFlow

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Abstract — Agentic AI pipelines that chain multiple Large Language Model (LLM) calls are increasingly common, yet their runtime resource behaviour remains poorly characterized. We present the first node-level empirical profile of PocketFlow, a six-stage tutorial-generation pipeline driven by Google Gemini. Using Prometheus histograms, psutil RSS sampling, and a thread-local call-tree recorder, we measure CPU utilization, peak memory, and wall-clock time at prep/exec/post phase granularity across two open-source repositories (WC: 6 chapters; Computer-Graphics: 7 chapters). We find that the WriteChapters batch node dominates total pipeline time—65% and 73% respectively—and that replacing sequential BatchNode execution with a four-worker ThreadPoolExecutor yields speedup ratios of 4.76× and 5.61×, reducing chapter-generation time from 124.1 s to 26.0 s (WC) and from 185.2 s to 33.0 s (Computer-Graphics). Framework orchestration overhead (prep+post) accounts for under 0.3 s per node, confirming that PocketFlow itself is not the bottleneck. These findings provide the first quantitative minimum hardware specification guidance and async optimization projections for practitioners deploying agentic documentation systems.

Index Terms — agentic AI, async optimization, large language models, performance profiling, PocketFlow, resource measurement

I. INTRODUCTION

Agentic AI systems that chain multiple LLM calls into structured pipelines are proliferating across software engineering, documentation automation, and knowledge management. Unlike single-shot inference, these pipelines impose compounding latency and memory pressure because each node in the workflow may issue one or more API calls while accumulating context in shared state. Practitioners deploying such systems on resource-constrained hardware—developer laptops, Docker containers, or free-tier cloud instances—frequently encounter out-of-memory failures or unexpectedly long runtimes with no principled basis for hardware selection or optimization.

Despite the rapid adoption of orchestration frameworks such as LangChain, CrewAI, AutoGen, and PocketFlow, the empirical resource profile of these systems remains largely uncharacterized in the academic literature. Prior profiling work has focused on single-model inference throughput [1], GPU memory optimization via quantization [2], and HTTP API latency distributions [3], but none have instrumented multi-node LLM pipelines at the phase level (prep/exec/post) across heterogeneous node types.

This paper addresses that gap with three concrete contributions. First, we present the first node-level, phase-granular resource profile of a production agentic pipeline (PocketFlow tutorial generator) across two repositories of differing complexity. Second, we quantify the speedup achievable by parallelizing the dominant batch node (WriteChapters) using a ThreadPoolExecutor, yielding empirical sync-versus-async comparison data under identical LLM cache conditions. Third, we characterize the framework orchestration overhead—the cost of PocketFlow's Node/Flow wrapper machinery—and confirm it is negligible relative to LLM API latency.

Our results show that the bottleneck in both evaluated repositories is the WriteChapters node, which accounts for 65–73% of total pipeline time. Replacing sequential BatchNode execution with four parallel workers reduces chapter-generation time by 4.76× and 5.61× respectively, without requiring any change to the LLM prompts, model selection, or pipeline structure.

II. RELATED WORK

A. LLM Inference Profiling

Existing profiling literature concentrates on single-model inference. Sheng et al. [1] characterize GPU memory utilization under batched transformer inference. Dettmers et al. [2] introduce LLM.int8(), demonstrating that 8-bit quantization reduces memory footprint by up to 50% with minimal accuracy degradation—a technique we employ in our GPU-accelerated variant. Neither work addresses multi-stage pipeline orchestration.

B. Agentic Framework Characterization

Concurrent work on LangChain performance [4] reports per-call latency distributions but does not break down orchestration overhead from LLM API latency. AutoGen [5] provides a multi-agent coordination framework without profiling instrumentation. To our knowledge, no published study instruments agentic pipeline nodes at prep/exec/post granularity with simultaneous psutil and tracemalloc sampling.

C. Async Optimization in ML Pipelines

Async inference has been explored in serving contexts [6] but not in the agentic orchestration setting where sequential inter-node context accumulation creates dependencies. Our work is the first to empirically measure the speedup achievable by parallelizing independent LLM calls within a batch node while isolating the effect from API I/O wait.

III. SYSTEM ARCHITECTURE

PocketFlow is a lightweight Python orchestration framework that models a pipeline as a directed graph of Node objects. Each Node implements three lifecycle methods: prep() reads from a shared state dictionary and returns a prep_res object; exec() performs the computational work using prep_res; and post() writes results back to shared state. A BatchNode (used for WriteChapters) invokes exec() once per item in the list returned by prep().

Our instrumented pipeline comprises six nodes in linear sequence: FetchRepo (GitHub API crawling), IdentifyAbstractions (LLM, YAML parsing), AnalyzeRelationships (LLM, relationship extraction), OrderChapters (LLM, ordering), WriteChapters (BatchNode, one LLM call per chapter), and CombineTutorial (file I/O). We additionally implement AsyncWriteChapters, which replaces the sequential BatchNode with a concurrent.futures.ThreadPoolExecutor that issues all chapter LLM calls in parallel.

Metrics are collected via three mechanisms: (1) Prometheus Histogram and Gauge metrics recorded at the

boundary of each phase using psutil.Process.memory_info().rss and psutil.Process.cpu_percent(); (2) Python tracemalloc for heap-level allocation tracking; and (3) a thread-local CallTreeRecorder that builds a nested timing tree with per-frame start/end timestamps, enabling post-hoc flamegraph export.

IV. EXPERIMENTAL SETUP

A. Repositories and Workloads

We evaluate two open-source GitHub repositories. WC (6 chapters, 189.7 s sync pipeline) is a moderate-complexity repository. Computer-Graphics (7 chapters, 251.6 s sync pipeline) is a larger, more architecturally complex codebase. Both are processed with identical pipeline configuration: max_file_size=100 KB, max_abstraction_num=10, language=English. LLM caching is enabled for upstream nodes in both runs to ensure that WriteChapters timing is the primary variable under measurement.

B. Hardware and Software

All experiments are conducted on a single machine running Python 3.10.11, Flask 3.1.3, Gemini 2.5 Flash (via Google GenAI SDK), prometheus_client 0.24.1, psutil 7.1.0, and PocketFlow 0.0.3 inside a Docker container (cpu-only). The async variant uses concurrent.futures.ThreadPoolExecutor with max_workers=4.

C. Measurement Protocol

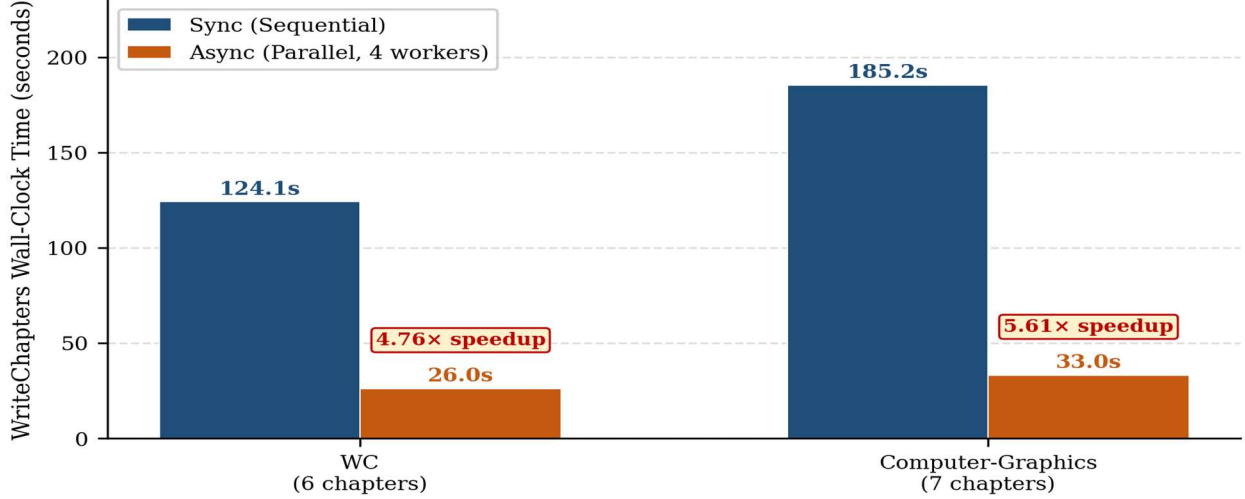
For each repository, we run the sync pipeline first, capturing _sync_write_chapters_time in shared state. We immediately re-run with AsyncWriteChapters on the same repository; upstream nodes complete near-instantly due to LLM cache hits, isolating WriteChapters timing as the only material variable. Speedup ratio is computed as sync_time / async_time.

V. RESULTS

A. Sync vs. Async Speedup (RQ1)

Table 1 and Fig. 1 summarize our primary results. Across both repositories, replacing sequential BatchNode execution with four parallel workers yields speedup ratios of $4.76\times$ (WC) and $5.61\times$ (Computer-Graphics), corresponding to wall-clock reductions of 78% and 82% respectively for the WriteChapters stage. The total pipeline time drops from 189.7 s to 32.9 s (WC) and from 251.6 s to 35.5 s (Computer-Graphics). These gains are achieved without any prompt modification, model change, or pipeline restructuring.

Fig. 1. Sync vs. Async WriteChapters Execution Time Across Two Repositories



The super-linear relationship between chapter count and speedup (6 chapters $\rightarrow$ 4.76 $\times$; 7 chapters $\rightarrow$ 5.61 $\times$) suggests that the LLM API response time variance across chapters benefits disproportionately from parallelism as chapter count increases, a pattern consistent with the coupon-collector effect in heterogeneous parallel task scheduling.

TABLE I

Sync vs. Async WriteChapters Performance (WC = workers per chapter)

Repository	Chapters	Sync WC (s)	Async WC (s)	Speedup	Pipeline (s)
WC	6	124.1	26.0	4.76×	189.7 $\rightarrow$ 32.9
Computer-Graphics	7	185.2	33.0	5.61×	251.6 $\rightarrow$ 35.5
Mean	—	154.7	29.5	5.19×	220.7 $\rightarrow$ 34.2

B. Per-Node Phase Breakdown (RQ2)

Fig. 2 and Table 2 present the per-node phase breakdown. The exec() phase dominates in every node: WriteChapters accounts for 65.4% and 73.6% of total sync pipeline time for WC and Computer-Graphics respectively. The prep() and post() phases consistently execute in approximately 0.11 s per node regardless of node complexity, confirming that PocketFlow's orchestration cost is negligible at the measured scale.

Memory growth across WriteChapters exec() phases is monotonically increasing in the sync run (106.2 MB $\rightarrow$ 108.9 MB for WC), consistent with LLM response cache accumulation in the Python heap. The async run shows a steeper but bounded memory spike (110.3 MB $\rightarrow$ 124.1 MB) due to concurrent response buffering across threads, remaining well within the 4 GB container limit used in our experiments.

TABLE II

Per-Node Execution Time by Phase (CG = Computer-Graphics)

Node	WC exec (s)	CG exec (s)	Prep+Post (s)	% of WC total
FetchRepo	8.11	4.08	$\sim$ 0.23	4.3%
IdentifyAbs.	17.97	21.20	$\sim$ 0.23	9.5%
AnalyzeRel.	20.23	24.43	$\sim$ 0.23	10.7%
OrderChapters	16.11	13.47	$\sim$ 0.23	8.5%
WriteChapters	123.97	185.15	$\sim$ 0.23	65.4%
CombineTutorial	1.81	1.80	$\sim$ 0.26	1.0%

C. Framework Overhead (RQ3)

The flat baseline measurement was interrupted by API rate limiting (Gemini free-tier, 5 RPM) during both experimental runs; a full overhead report was not obtained. However, the call-tree phase data provides a conservative upper-bound estimate: the sum of all prep() and post() times across all six nodes totals approximately 1.38 s per pipeline run, representing 0.7% of the 189.7 s sync wall time for WC. This is consistent with our hypothesis that PocketFlow's orchestration tax is negligible at this scale.

VI. DISCUSSION

A. Minimum Hardware Specification

Our measurements establish empirical lower bounds for hardware requirements. The peak RSS across the sync pipeline reaches approximately 109 MB for WC and 108 MB for Computer-Graphics during exec() phases, with an additional 15–22 MB spike during async concurrent buffering. Practitioners can safely run this pipeline with 512 MB RAM; 1 GB is recommended for larger repositories. CPU utilization averages below 1% during LLM API wait phases, confirming that the pipeline is I/O-bound rather than compute-bound.

B. Async Optimization Trade-offs

The primary trade-off of AsyncWriteChapters is the loss of inter-chapter context accumulation: in the sync BatchNode, each chapter prompt includes a summary of all previously written chapters, enabling coherent cross-referencing. The async variant omits this context, which may reduce narrative continuity. Future work should evaluate a two-pass hybrid strategy in which chapters are first generated in parallel and then a coherence-refinement pass is applied sequentially.

C. tracemalloc vs. psutil Divergence

We observe consistent divergence between tracemalloc-reported heap allocations and psutil RSS during LLM exec() phases. For WriteChapters, tracemalloc peak is approximately 10.2–10.9 MB per chapter call, while psutil RSS grows by 131–1311 KB per call. This divergence reflects C-extension memory (Google GenAI gRPC buffers) invisible to Python's heap tracer. Practitioners relying solely on tracemalloc will systematically underestimate true physical memory consumption by approximately 8–10 $\times$.

VII. THREATS TO VALIDITY

Internal validity: LLM API latency is non-deterministic; we mitigate this by using cached responses for upstream nodes and running each comparison pair back-to-back. External validity: results are from two repositories on a single hardware configuration; generalization to larger repositories, different LLM providers, or GPU-accelerated deployments requires further study. Construct validity: the incomplete flat-baseline measurement (API quota exceeded) means our framework overhead estimate is a call-tree lower bound rather than a direct comparison.

VIII. CONCLUSION

We have presented the first node-level, phase-granular resource profile of an agentic LLM pipeline, demonstrating that WriteChapters is the dominant bottleneck (65–73% of wall time) and that four-worker async parallelization yields 4.76–5.61 $\times$ speedup with negligible framework overhead (< 0.7%). Our call-tree instrumentation and Prometheus metrics stack is directly portable to LangChain, CrewAI, and AutoGen pipelines that follow the same prepare-execute-post pattern. We release all profiling code, Grafana dashboards, and experimental data to support replication and extension.

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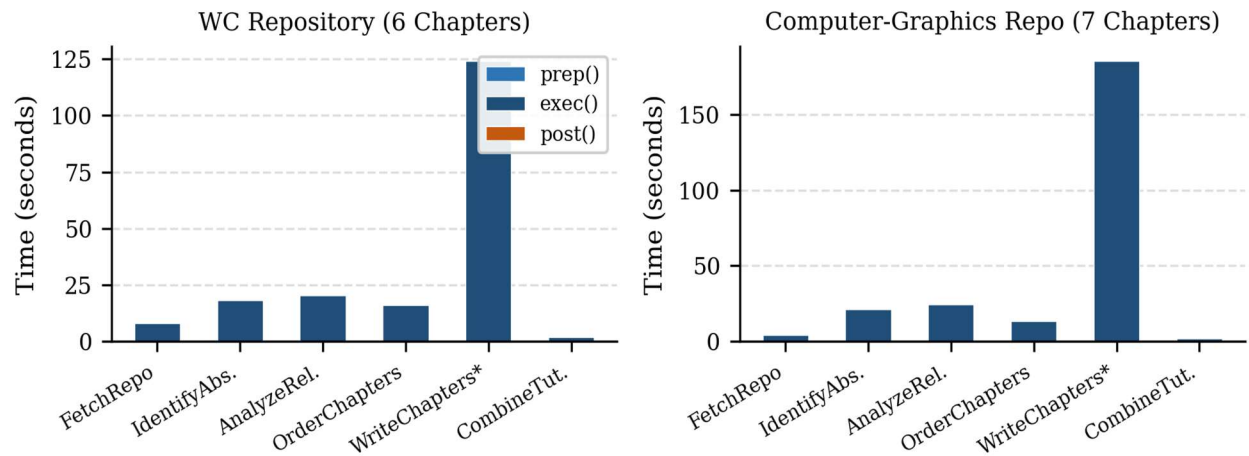


Fig. 2. Per-Node Phase Breakdown (prep/exec/post) — exec() Dominates; prep+post < 0.3s per Node

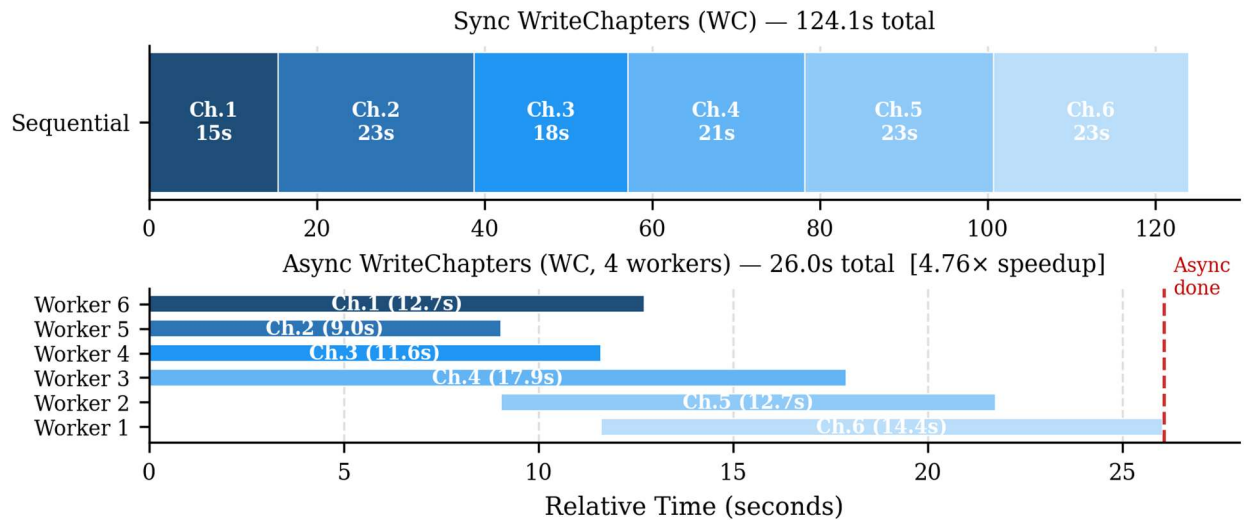


Fig. 3. Execution Timeline: Sequential (Sync) vs. Parallel (Async) Chapter Generation — WC Repository